

Project 2 - Econ 144

Eli Motzkin, Mason Lee

2025-11-10

Intro

For this project, we will be taking a look at the 30 year expected inflation rate. The data was collected monthly starting in the 80s, but we will only be using data from the past 15 years in order to stay consistent with the other dataset (30 year breakeven inflation rate) for the purpose of the VAR. The 30 year inflation rate is a FRED estimate for the average inflation rate across the next 30 years, and features trend, seasonality, and cycles.

Results

Packages

```
library(readr)
library(lattice)
library(foreign)
library(MASS)
library(car)
```

```
## Loading required package: carData
```

```
require(stats)
require(stats4)
```

```
## Loading required package: stats4
```

```
library(KernSmooth)
```

```
## KernSmooth 2.23 loaded
## Copyright M. P. Wand 1997-2009
```

```
library(fastICA)
library(cluster)
library(leaps)
library(mgcv)
```

```
## Loading required package: nlme
```

```
## This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```
library(rpart)
library(pan)
library(mgcv)
library(DAAG)
```

```
##
## Attaching package: 'DAAG'
```

```
## The following object is masked from 'package:car':
##
##      vif
```

```
## The following object is masked from 'package:MASS':
##
##      hills
```

```
library("TTR")
library(tis)
```

```
##
## Attaching package: 'tis'
```

```
## The following object is masked from 'package:TTR':
##
##      lags
```

```
## The following object is masked from 'package:mgcv':
##
##      ti
```

```
require("datasets")
require(graphics)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##      method      from
##      as.zoo.data.frame zoo
```

```
##
## Attaching package: 'forecast'
```

```
## The following object is masked from 'package:tis':
##
##      easter
```

```
## The following object is masked from 'package:nlme':
##
##      getResponse
```

```
require(astsa)
```

```
## Loading required package: astsa
```

```
##
```

```
## Attaching package: 'astsa'
```

```
## The following object is masked from 'package:forecast':
```

```
##
```

```
##      gas
```

```
library(RColorBrewer)
```

```
library(plotrix)
```

```
library(tsibble)
```

```
## Registered S3 method overwritten by 'tsibble':
```

```
##   method              from
```

```
##   as_tibble.grouped_df dplyr
```

```
##
```

```
## Attaching package: 'tsibble'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, union
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
library(USgas)
```

```
library(fpp3)
```

```
## -- Attaching packages ----- fpp3 1.0.2 --
```

```
## v tibble      3.2.1      v tsibbledata 0.4.1
```

```
## v dplyr       1.1.4      v feasts      0.4.2
```

```
## v lubridate   1.9.4      v fable       0.4.1
```

```
## -- Conflicts ----- fpp3_conflicts --
```

```
## x feasts::ACF()          masks nlme::ACF()
```

```
## x dplyr::between()       masks tis::between()
```

```
## x dplyr::collapse()      masks nlme::collapse()
```

```
## x lubridate::date()      masks base::date()
```

```
## x lubridate::day()       masks tis::day()
```

```
## x dplyr::filter()        masks stats::filter()
```

```
## x lubridate::hms()       masks tis::hms()
```

```
## x tsibble::intersect()   masks base::intersect()
```

```
## x lubridate::interval()  masks tsibble::interval()
```

```
## x dplyr::lag()           masks stats::lag()
```

```
## x lubridate::month()     masks tis::month()
```

```
## x lubridate::period()    masks tis::period()
```

```
## x lubridate::POSIXct() masks tis::POSIXct()
## x lubridate::quarter() masks tis::quarter()
## x dplyr::recode() masks car::recode()
## x dplyr::select() masks MASS::select()
## x tsibble::setdiff() masks base::setdiff()
## x lubridate::today() masks tis::today()
## x tsibble::union() masks base::union()
## x lubridate::year() masks tis::year()
## x lubridate::ymd() masks tis::ymd()
```

Data

```
data <- read_csv("30expected.csv")
```

```
## Rows: 189 Columns: 2
## -- Column specification -----
## Delimiter: ","
## dbl (1): EXPINF30YR
## date (1): observation_date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

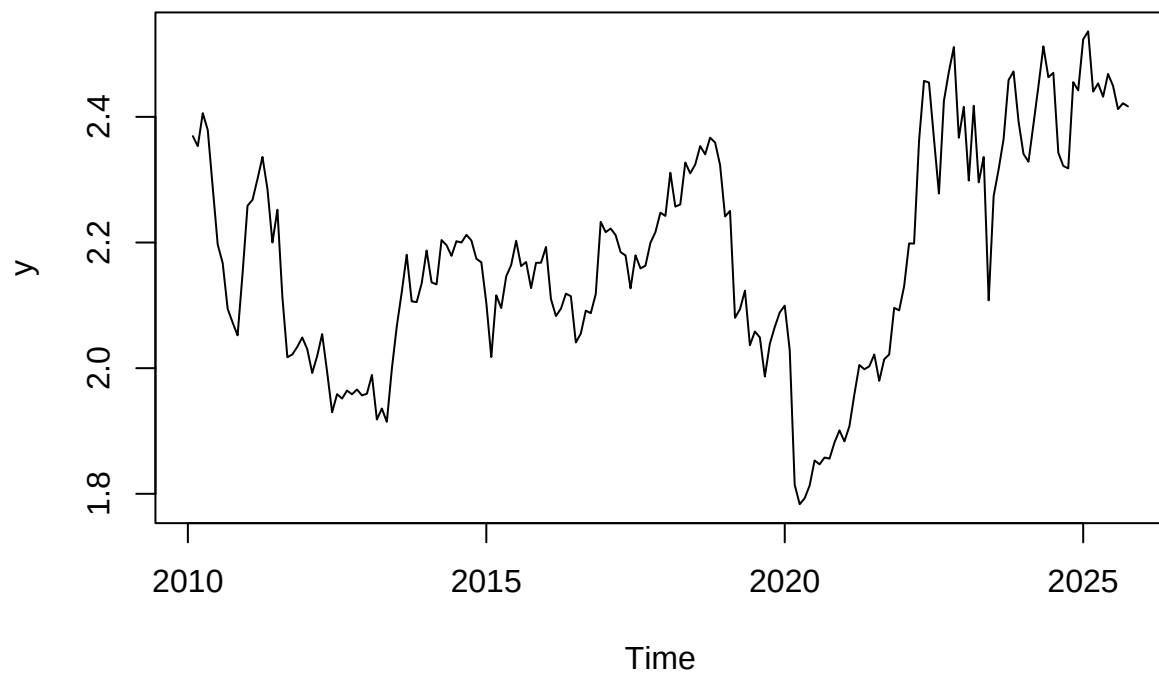
```
head(data)
```

```
## # A tibble: 6 x 2
##   observation_date EXPINF30YR
##   <date>          <dbl>
## 1 2010-02-01      2.37
## 2 2010-03-01      2.35
## 3 2010-04-01      2.41
## 4 2010-05-01      2.38
## 5 2010-06-01      2.29
## 6 2010-07-01      2.20
```

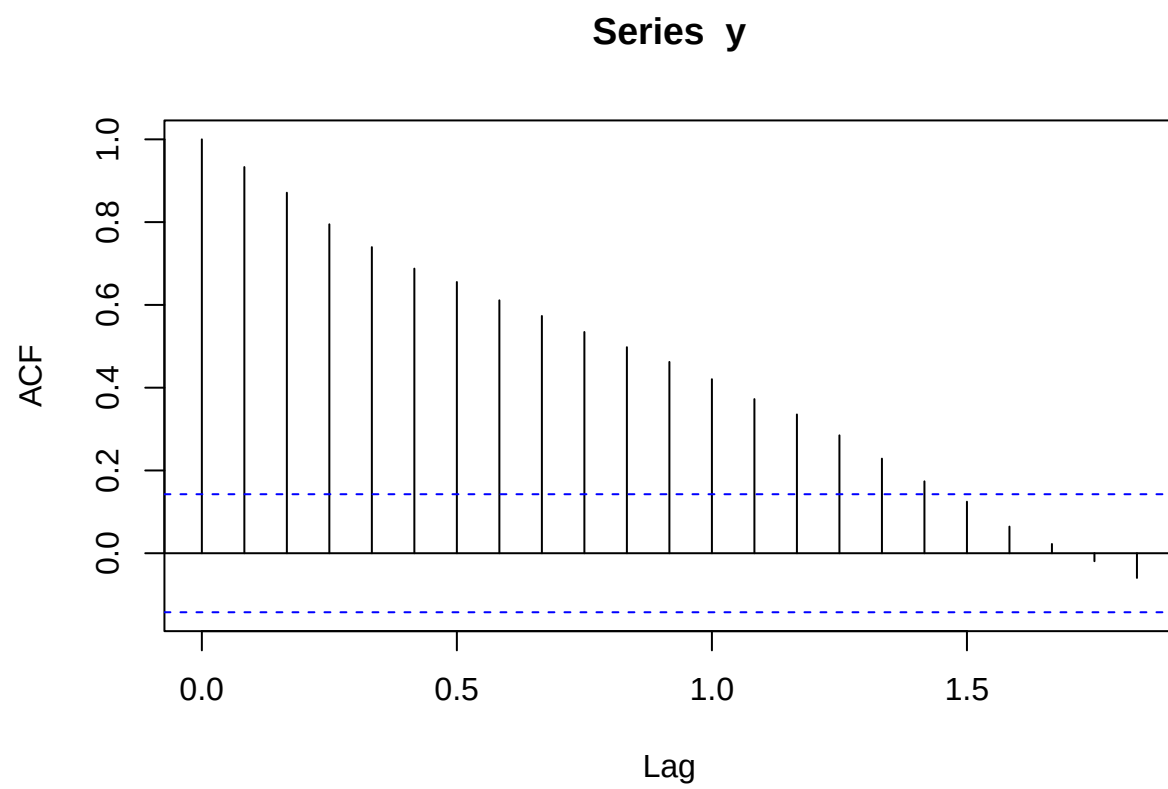
Part A: Time Series Plot

```
y <- ts(
  data$EXPINF30YR,
  start = c(year(data$observation_date[1]), month(data$observation_date[1])),
  frequency = 12
)

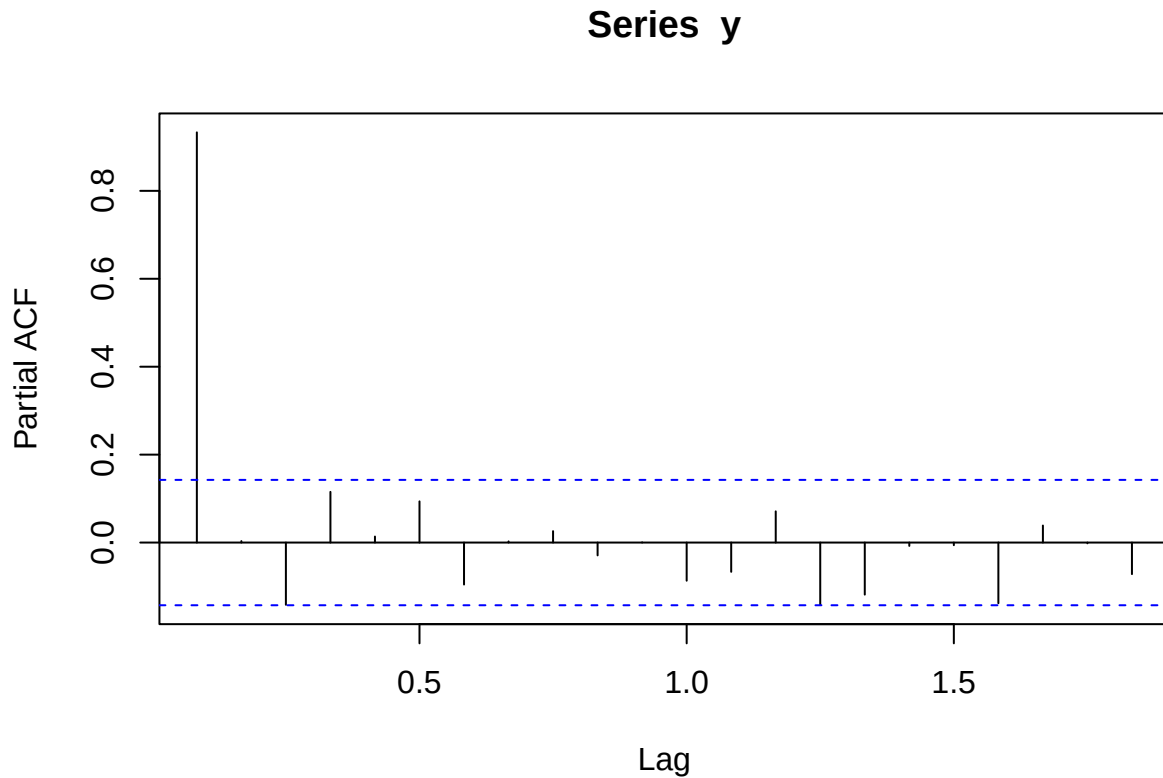
plot(y)
```



`acf(y)`



```
pacf(y)
```

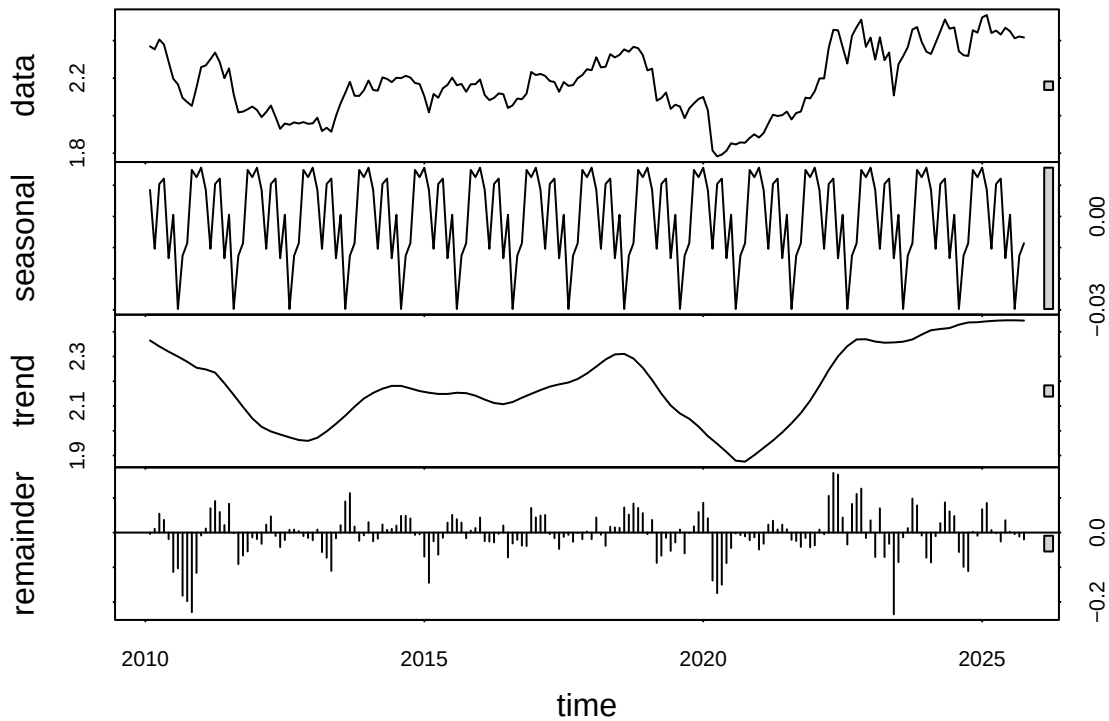


Above, we see a time series plot of the 30 year expected inflation rate. It shows a slight upward trend as the market recovered from the 2008 recession, and plenty of seasonality. There are also longer term fluctuations (cycles) present.

The ACF and PACF plot are a textbook example of an AR(1) pattern. There is 1 strong PACF spike, and the ACF plot shows a slow regression towards 0. This means that 30 year projections are heavily dependent on the most recent months projection.

Part B: STL Decomposition

```
fit <- stl(y, s.window = "periodic", robust = TRUE)
plot(fit)
```



The STL decomposition confirms that this dataset has trend, seasonality, and cycles. The seasonal decomp shows a steady, rhythmic pattern, and the remainder shows that there are cycles present. The trend is slight, but definitely present. The data starts on a slow downward trend and then rebounds before crashing in 2020 (Covid-19). After the pandemic, the trend returned to its upward climb.

Part C: Model Fitting

```
t <- seq_along(y)
Sd <- seasonaldummy(y)

fit <- Arima(y, order = c(1,0,0), seasonal = c(0,0,0),
            xreg = cbind(trend = t, Sd),
            include.mean = FALSE)

summary(fit)
```

```
## Series: y
## Regression with ARIMA(1,0,0) errors
##
## Coefficients:

## Warning in sqrt(diag(x$var.coef)): NaNs produced

##      ar1    trend    Jan    Feb    Mar    Apr    May    Jun    Jul
```



```
##          1  0.0072  0.0071 -0.0082 -0.0259 -0.0057  0.0053 -0.0286 -0.0129
## s.e.   NaN  0.0043  0.0142  0.0191  0.0220  0.0239  0.0249  0.0252  0.0249
##          Aug      Sep      Oct      Nov
##        -0.0391 -0.0323 -0.0268 -0.0022
## s.e.    0.0239  0.0220  0.0191  0.0142
##
## sigma^2 = 0.00357:  log likelihood = 270.64
## AIC=-513.28  AICc=-510.86  BIC=-467.89
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set -0.006713912 0.05765482 0.0422289 -0.3466048 1.935959 0.2965048
##              ACF1
## Training set -0.01175118
```

The above model fits 12 dummy seasonal variables, trend, and AR(1) cyclical behavior together into a model called “fit”. AR(1) comes from order = c(1,0,0) in the Arima function and was chosen by the ACF and PACF spikes. The seasonal trend (s = 12) is due to how the data was collected. The deterministic trend component (t) is also included in the model and shows a general positive correlation. This oversimplifies things given the time series graph, but does add to the model.

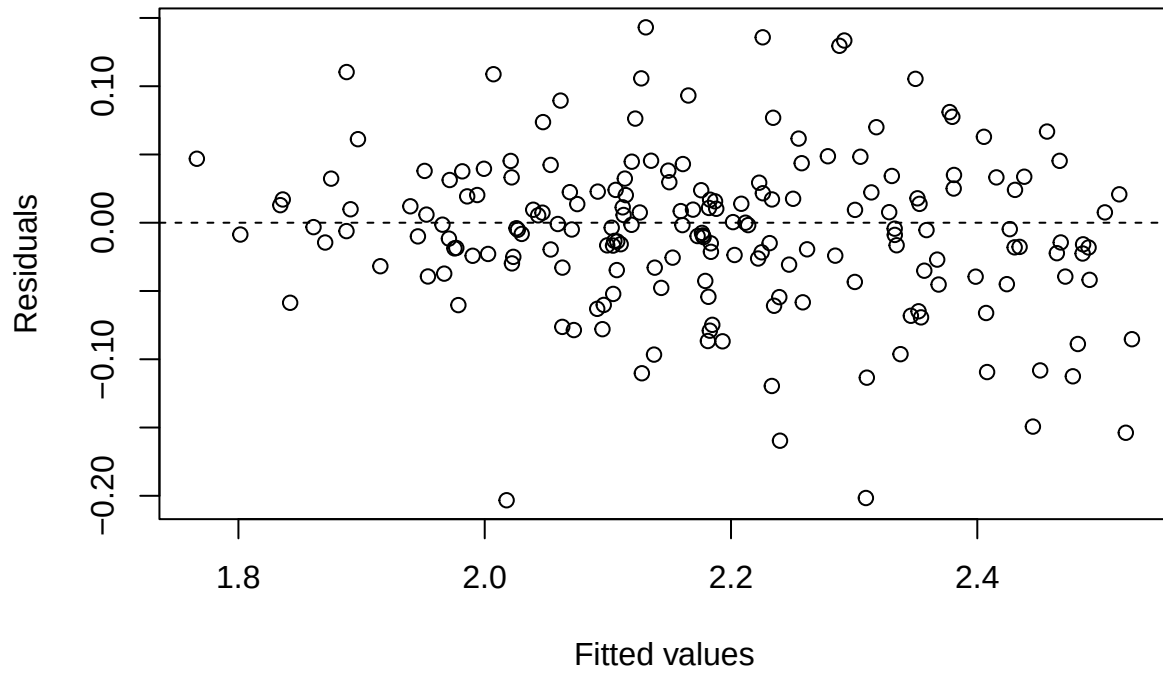
The AIC of the model is -513 and BIC is -467, which doesn’t tell us much yet but will be helpful when we start comparing different models.

Part E: Residuals vs. Fitted

```
fitted <- fitted(fit)
resid <- residuals(fit)

plot(fitted, resid,
     xlab = "Fitted values",
     ylab = "Residuals",
     main = "Residuals vs Fitted (Part C model)")
abline(h = 0, lty = 2)
```

Residuals vs Fitted (Part C model)



The Residual vs. Fitted plot shows a very random collection of points, which is exactly what we hope to see. There is some heteroskedasticity, which isn't ideal, but it is clear that our model is on the right path.